ARASH VASHAGH

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Summary

Machine Learning Developer with research publications and industry certifications in AI, cybersecurity, and cloud. Experienced in building robust ML pipelines and adversarially resilient models using PyTorch, with hands-on experience in security tools and cloud platforms (AWS and Azure). Strong at translating research into practical solutions.

Skills

Programming: Python, Java, JavaFX, Kotlin, SQL, C++
Backend & APIs: REST API integration, JSON processing
Frameworks & Libraries: PyTorch, TensorFlow, Flask, FastAPI, Scikit-learn
Cloud Platforms: AWS, Microsoft Azure
Tools & Environments: Git, Docker, IntelliJ IDEA, VS Code, Jupyter, Wireshark
Languages: Persian (Native), English (Professional)

Certifications

AI and Data Analysis
- AWS Certified AI Practitioner (Active: Jul 2025 – Jul 2028)
- CompTIA Data+ (DA0-001) (Active: Aug 2025 – Aug 2028)
- Microsoft Certified: Azure AI Fundamentals (AI-900) (Earned: Jul 2025)
- Microsoft Certified: Azure Data Fundamentals (DP-900) (Earned: Jul 2025)

Cloud Computing
- AWS Certified Cloud Practitioner (Active: Jun 2025 – Jun 2028)
- Microsoft Certified: Azure Fundamentals (AZ-900) (Earned: Aug 2025)

Cybersecurity and Database Systems
- ISC2 SSCP (In Progress, Expected September 2026)
- CompTIA Security+ (SY0-701) (Active: Sep 2025 – Sep 2028)
- Microsoft Certified: Security, Compliance & Identity Fundamentals (SC-900) (Earned: Jul 2025)
- CompTIA DataSys+ (Beta Exam Completed: Mar 2026 – Results Pending)

Project Management
- CompTIA Project Management Essentials (Issued: Jul 2025)

Experience

Team Lead — Trustworthy and Secure AI (TSAI) Lab — Fredericton, Canada **Oct 2025 – Present**
- Overseeing lab operations and coordinating graduate student work.
- Reporting progress and operational updates to the supervising faculty member.
- Managing the TSAI LinkedIn page for lab communications and educational content.

Graduate Research Assistant — University of New Brunswick (UNB) — Fredericton, Canada **Sep 2024 – Present**
- MSc thesis research in Adversarial Machine Learning (robustness and attack detection).
- Co-authored a survey proposing a unified framework to compare adversarial attacks across ML and XAI.
- Multiple manuscripts under review at top journals and conferences. Additional works in progress.

Graduate Teaching Assistant — UNB — Fredericton, Canada **Jan 2025 – Apr 2026**
- *Software Engineering* (Winter 2026) – Java
- *Data Communication and Network Modeling* (Fall 2025) – Python
- *Data Science for Big Data Analytics* (Winter 2025) – Python

Conference Reviewer **Jul 2024 – Oct 2024**

- Reviewed manuscripts and provided structured feedback at international conferences:
  - *ArtInHCI 2024* – Artificial Intelligence and Human–Computer Interaction (Kunming, China)
  - *BIBE 2024* – Biological Information and Biomedical Engineering (Hohhot, China)
- Awarded official reviewer certificates.

Research Assistant — Isfahan University of Technology (IUT) — Isfahan, Iran **Feb 2021 – Feb 2024**

- Conducted research on machine learning and healthcare, resulting in two IEEE publications.
- Collaborated with faculty on model design, data preprocessing, and experimental evaluation.
- Supported technical writing and presentation preparation for joint research projects.

Teaching Assistant — IUT — Isfahan, Iran **Feb 2021 – Feb 2024**

- Mentored over 400 students in undergraduate courses including Artificial Intelligence, Embedded Systems, Data Structures, and Programming Labs in C, C++, Java, and Python.
- Led tutorials, graded assignments, and guided students through programming and algorithmic exercises.

Education

Master's Candidate — University of New Brunswick (UNB), Fredericton, Canada **Sep 2024 – Aug 2026**

- Awarded Graduate Scholarship and Research Grant.
- Major: Computer Science
- Research Focus: Adversarial Machine Learning

Bachelor of Science — Isfahan University of Technology (IUT), Isfahan, Iran **Sep 2019 – Feb 2024**

- Major: Computer Engineering
- Minor: Intelligent Systems

Selected Projects

PromptShield: Adversarial Detection for Vision-Language Models **Jan 2026 - May 2026**

- Developed a novel framework to detect adversarial images in vision-language models.
- Evaluated the framework across multiple datasets, model backbones, and adversarial attacks.
- Technologies: Python, PyTorch, OpenCLIP, NumPy, Scikit-learn.
- Manuscript and code under review at NeurIPS 2026.
- Source code and complete description available at GitHub.

AttackBench: Adversarial Attack Visualization Tool **Mar 2026 – Apr 2026**

- Developed a desktop application to generate and visualize adversarial examples on image classification models.
- Implemented common adversarial attacks and displayed their impact on model predictions and confidence.
- Built a Java-based UI with a Python backend to load models, run attacks, and return results in real time.
- Technologies: Java (JavaFX), Python, PyTorch, FastAPI.
- Source code and complete description available at GitHub.

PhishGuard: Phishing URL Detection App **Mar 2026 – Apr 2026**

- Developed an Android application using Kotlin with a backend service for phishing URL detection.
- Integrated RESTful API communication between mobile client and Python backend.
- Implemented input validation and security checks including URL formatting, DNS lookup, and SSL verification.
- Provides predictions with confidence scores and user-friendly feedback.
- Technologies: Kotlin, Python, Flask, Scikit-learn, Retrofit.
- Source code and complete description available at GitHub.